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Structure of High-Pressure Premixed Ammonia, Hydrogen, and Methane Flames With 100 kHz OH-PLIF Measurements

The impact of fuel composition on flame dynamics is investigated at 10 bar using 100 kHz hydroxyl planar laser induced fluorescence (OH-PLIF) measurements. The PLIF images are synchronized with high-frequency pressure measurements at distinct locations in the chamber to investigate mechanisms for thermo-acoustic feedback with the flame heat release and resonant acoustic modes of the combustor. Experiments were conducted over a range of fuel compositions spanning a hydrogen fuel fraction (χ) of 30% to 90%, and an ammonia decomposition efficiency (η) from 40% to 100%. In predominantly hydrogen-rich fuel mixtures ($\chi, \eta > 0.8$), flame stabilization occurs along the shear layer, and flame holding is observed closer to the burner face. As methane composition increases ($\chi < 0.8$), significant flame corrugation and commensurate high heat release rate are observed as chemical kinetic rates and thermodiffusive effects decrease. In these cases, OH radical production and transport between neighboring injector elements are observed. The flame heat release and transport of combustion products in this region occur during the compression phase of the acoustic cycle and are responsible for ignition of fresh reactants. These periodic ignition, heat release, and flame displacement at the resonant acoustic mode frequency of the combustor result in self-excited longitudinal combustion instabilities ($p'_{RMS}/p_{CC} \sim 1$–7%). With significant concentrations of methane and ammonia in the fuel blend ($\chi < 0.9, \eta < 0.6$), the amplitude of this instability increases ($p'_{RMS}/p_{CC} > 7\%$), and periodic global extinction and re-ignition of flames are observed with acoustic expansion and compression.
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Keywords: combustion instability, hydrogen combustion, ammonia combustion, high-pressure flames, lean-premixed flames, OH-PLIF

1 Introduction

Concerns of increasing global greenhouse gas emissions have sparked rapid development efforts in carbon-free combustion technology [1]. In stationary power generation, fuels such as hydrogen (H_2) and ammonia (NH_3) have potential as carbon-free alternatives to natural gas but present unique challenges that inhibit widespread utilization. Hydrogen has established methods for renewable production and possesses the highest gravimetric energy density of any fuel. However, its low volumetric energy density makes it prohibitive for efficient storage, transport, and distribution at an industrial scale. The availability and cost of hydrogen derived from carbon-free sources also remain challenging, especially at the scale necessary to replace natural gas. Hydrogen also presents several operability concerns as a drop-in alternative to natural gas. The flame speed of hydrogen is an order of magnitude greater than that of methane, the primary constituent of natural gas. The large increase in kinetic timescale of hydrogen flames leads to an increased risk of flashback [2]. Ammonia, on the other hand, contains a larger density of hydrogen atoms than liquid hydrogen. Pre-existing facilities for the production, storage, and transport of ammonia can also reduce the economic complexity of the fuel's large-scale implementation [3]. However, the slow kinetic rates of

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pure ammonia combustion limit the operating space of retrofit gas turbine combustors [4]. Moreover, the nitrogen atom present in the fuel introduces unique chemical pathways which can lead to high NO_x emissions [5]. Blending of ammonia with hydrogen can mitigate undesirable properties of the individual fuels [6]. Nevertheless, carbon-free developments in combustor technology must accommodate these fuel blends with natural gas over a wide range of compositions to offset discrepancies in energy demand and fuel availability.

Lean-premixed combustion is common for low NO_x natural gas-fired power generation gas turbine combustors. Successful retrofitting of these combustors to operate with hydrogen and ammonia fuel blends requires understanding of the physical processes that govern the structure and propagation of the flame. Premixed hydrogen combustion has distinctly different underlying physics governing flame propagation compared to methane [7–10]. One such difference is the response to flame stretch, a phenomenon often described by the Markstein length. The Markstein length is a characteristic length scale that quantifies the sensitivity of local flame speed to flame curvature [11]. The Markstein number is a nondimensional parameter defined as the ratio of Markstein length to flame thickness. At lean conditions, Al-Khafaji et al. [12] reported a decrease in Markstein number and subsequent increase in stretch sensitivity with hydrogen addition to methane flames. For ammonia–hydrogen fuel blends, Ichikawa et al. [9] noted a nonmonotonic trend of Markstein length with variation in hydrogen mole fraction. The effect of hydrogen addition in lean ammonia flames is less understood, especially at engine relevant conditions. At elevated pressure, the operability space of these fuel mixtures has been characterized for several injector designs [4,13]. However, these studies are also scarce, and further experimental efforts at high pressure are necessary to resolve the fundamental processes affecting the structure and dynamics of these flames.

Despite the lack of empirical data, first principles analysis can still provide insight into the effect of hydrogen addition on the propagation of ammonia flames. Incommensurate rates of thermal and mass diffusion (nonunity Lewis number) have a prominent effect on flame stretch sensitivity [14]. When the Lewis number is less than one, typical of hydrogen combustion, mass diffusion occurs at a faster rate than thermal diffusion [15], leading to a thermodiffusive mechanism for intrinsic flame instabilities to develop [16]. In contrast, Lewis numbers closer to unity are typical for methane combustion, and preferential thermodiffusive instabilities are less likely to occur. In premixed turbulent flames, this effect is believed to greatly increase the flame speed as turbulent intensity enhances wrinkling and leads to the development of smaller-scale structures in the flame [10,17,18]. A recent numerical study by Gaucherand et al. [8] investigated this effect on H_2/NH_3 flames and determined that ammonia addition led to reduced stretch sensitivity for the examined operating conditions.

Lean-premixed combustors are notably susceptible to combustion instabilities due to an increased sensitivity of flame thickness, flame speed, and reaction rates to variations in equivalence ratio near the lean blowout limit [19–23]. Thermo-acoustic instabilities result from in-phase coupling of heat release and acoustic pressure fluctuations. Several intermediary coupling mechanisms exist to close the feedback loop between pressure oscillations and heat release. In technically premixed combustors, differences in acoustic impedance between the locations of fuel and air introduction can cause pressure fluctuations to perturb local equivalence ratio and thus heat release [24]. Velocity fluctuations, be it from acoustic disturbances or turbulent eddies, lead to fluctuations in flame area and ensuing heat release. These flame area distortions can result in different timescales of heat release depending on the fuel composition. Stretch sensitive flames can see rapid annihilation of flame curvature through kinematic restoration [25]. Over an acoustic period, this acts to broaden the flame heat release in the frequency domain, which can reduce the coherent response to acoustic fluctuations [26].

Characterization of the underlying physical mechanisms that result in significant changes to flame structure with small changes in fuel composition requires spatiotemporally resolved measurements of heat release. Laser induced fluorescence is a nonintrusive diagnostic capable of detecting key combustion radicals linked to heat release in the flame. Planar laser induced fluorescence of the hydroxyl radical (OH-PLIF) is one such application that has been previously employed in hydrogen flames [27–30]. Lantz et al. [27] performed the diagnostic at 10 Hz in two industrial gas turbine combustors and observed a reduction in flame front perturbations with hydrogen addition to methane. In radially stratified NH_3/H_2 flames, Jin and Kim [29] used OH-PLIF in conjunction with OH^*/NH_2^* chemiluminescence to identify disparate flame structures and timescales. They noted that the interaction between the kinetically slow ammonia flame zones and much faster hydrogen flames prevented the development of coherent flame undulations that can couple with acoustic perturbations. Despite the advantages of OH-PLIF, repetition rate remains a key practical limitation. Care is also necessary in the interpretation of planar measurements in a turbulent three-dimensional flow-field. Investigation of the structure and dynamics of these flames in response to acoustic disturbance requires deployment of the technique at rates capable of resolving the timescales of interest.

The effect of fuel composition on the structure and propagation of self-excited high-pressure ammonia/hydrogen/methane lean-premixed flames is studied in the current work. The directional sensitivity of combustion dynamics to fuel composition has been characterized in previous work utilizing this combustor [13]. An increase in pressure fluctuation amplitude (p'_{RMS}) with natural gas addition was observed. The frequency corresponded to a longitudinal full-wave (2L) mode of the combustion chamber. In the absence of natural gas, the instabilities were lower in amplitude and largely insensitive to ammonia addition. Ammonia addition with the inclusion of natural gas however resulted in increased p'_{RMS}. A mode transition from 1100 Hz to 550 Hz was also observed for sufficient ammonia addition to these fuel mixtures. This frequency corresponds to the fundamental half-wave (1L) longitudinal mode of the combustion chamber. Changes in turbulent flame structure and its response to acoustic disturbances are characterized with 100 kHz OH-PLIF measurements. The novel application of this diagnostic at this repetition rate, at elevated pressure, and with $NH_3/H_2/CH_4$ fuel blends facilitates analysis over a broader range of timescales relevant to unsteady heat release. Oscillations at frequencies associated with the longitudinal modes of the combustor are fully resolved. Additionally, further resolution of element to element interaction and small scale flame corrugation reveals how small scale structures respond to global combustor dynamics.

2 Experiment Description

2.1 Combustor. A multi-element premixed combustor operating at a mean chamber pressure (p_{CC}) of $\sim$10.8 bar is used in the study. Figure 1 depicts a schematic of the experiment. The combustor uses heated unvitiated air with a fuel blend of ammonia, methane, hydrogen, and nitrogen. A plenum supplies uniform air to each micromixer element through a choked orifice to isolate the air supply from pressure fluctuations in the combustion chamber. The micromixer is comprised of three segments, one each for ammonia, natural gas, and hydrogen injection. Combustion chamber and micromixer lengths are designed to promote self-excited longitudinal combustion instabilities in the combustor [20]. The 19 micromixer elements are distributed in a uniform hexagonal array. The separation between neighboring elements is twice the 6.22 mm element diameter. A fixed area sonic nozzle located at the end of the combustion chamber pressurizes the combustor. Critical-flow sonic venturi nozzles are close-coupled to the experiment and are used to meter the flowrate of all reactants [31]. Hydrogen and methane are supplied to the experiment from bulk storage at 400 bar. Liquid anhydrous ammonia is pressurized to target conditions in a piston accumulator assembly and vaporized for gaseous supply through a 54 kW electrical heater. Typical test durations are 1 s, during which all flow conditions and geometric parameters are maintained within

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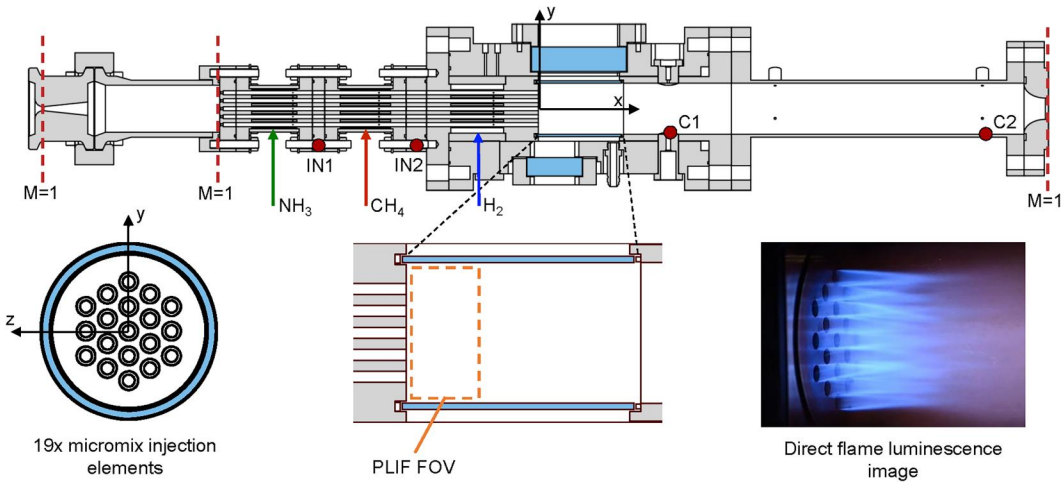

Fig. 1 Schematic of the combustor with instrumentation probe locations, OH-PLIF field of view, and representative image of flame chemiluminescence. Dashed vertical lines indicate regions of choked flow within the combustor. IN1, IN2, C1, and C2 depict the axial locations of high-frequency (2.5 MHz) pressure measurements.

an equivalence ratio uncertainty less than 2% of the mean. This duration is sufficient for the acquisition of steady, limit-cycle OH-PLIF and pressure measurements with negligible heat loss effects.

A parameterized expression of fuel composition is described in the following equation:

$$\chi\left[\eta\left(\frac{3}{2}H_2+\frac{1}{2}N_2\right)+(1-\eta)NH_3\right]+(1-\chi)CH_4 \tag{1}$$

The hydrogen fuel used in the experiment is assumed to be derived from ammonia as a hydrogen carrier through a decomposition process that yields a three to one molar ratio of hydrogen to nitrogen. The fraction of this mixture in the fuel is denoted by χ, where $\chi=0$ corresponds to operation with only CH_4 and $\chi=1$ corresponds to operation with only H_2/N_2. The fraction of hydrogen to ammonia is denoted by a decomposition efficiency η. For all test cases reported, the maximum computed χ uncertainty is 2.5%, and the maximum computed η uncertainty is 1.1%.

2.2 Diagnostics. Pressure fluctuations in the combustor were measured using recess mounted water-cooled high-frequency piezoresistive transducers (Kulite WCT312M-35/70BARA) at multiple axial locations in the combustor and sampled at 2.5 MHz [32]. The locations of the pressure transducers in the combustion chamber are marked with C1 and C2 in Fig. 1.

The OH-PLIF system used in this work is depicted in Fig. 2. The third harmonic (355 nm) from an Nd:YAG burst-mode laser is utilized to pump an optical parametric oscillator (OPO) in a double pass configuration (Spectral Energies QuasiModo, Beavercreak, OH). The OPO cavity is injection seeded with an ~850 nm external cavity diode laser (Sacher, Lion, Marburg, Germany) to narrow the ~609 nm signal line-width. This signal is mixed with the second harmonic output of the burst-mode laser (532 nm) to produce a system output frequency of 284 nm to target the $Q_2(8)$ line of the $A^2\Sigma^+(\nu'=1)\leftarrow X^2\Pi(\nu''=0)$ transition of OH. The system output pulse energy is ~1 mJ/pulse.

The 284 nm beam is collimated to a 42 mm tall sheet using a negative–positive cylindrical lens pair. The sheet is focused into the test section with an additional pair of $f=150$ mm cylindrical lenses. OH-PLIF signal is spectrally filtered with a 320/40 nm bandpass filter (Semrock 320/40 Brightline Bandpass) and collected with an $f=100$ mm $f/2.8$ UV grade objective lens (Cerco). Signal amplification and temporal gating is achieved using a dual-stage multichannel plate image intensifier (HiCATT 25 S20). Signal is collected over a 40 ns gate width for each image. Images are recorded on a Phantom TMX-7510 at 100 kHz with a 512 × 1280 pixel resolution. A 43 × 107 mm field of view is imaged, resulting in a spatial resolution of 83 μm/px. The exposure and gain settings on all cameras and intensifiers are consistent to ensure reliable comparison of the images.

To characterize the laser sheet profile, a UV beam sampler (Thorlabs BSF20-UV) is located immediately downfield of the first focusing lens and reflects ~ 8% of the incident laser sheet onto an N-BK7 glass diffuser (Thorlabs DG20-600-MD). Fluorescence from the glass is imaged on a Phantom v411 at 100 kHz with a 64 × 512 pixel resolution. A neutral density filter of OD 0.1 is installed on this camera to reduce signal intensity and prevent saturation. Images from the Phantom v411 are recorded simultaneously with the TMX-7510, so that every PLIF image has an accompanying image of the laser sheet. Figure 3(*a*) illustrates the summed laser intensity over a 3.5 ms burst duration. Due to the noticeable decay in laser intensity

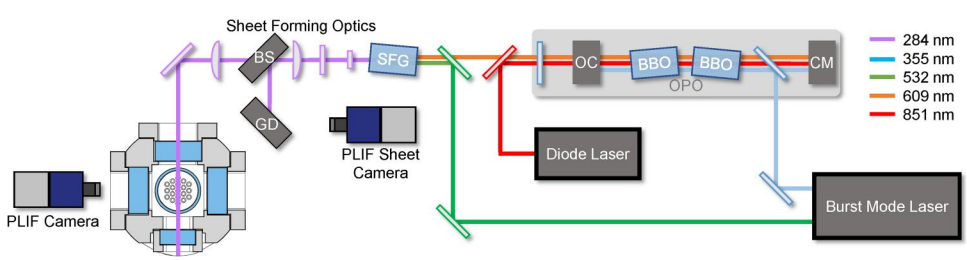

Fig. 2 Diagram of the OH-PLIF experimental configuration. CM: cavity mirror; BBO: beta-barium borate crystal; OC: output coupler; SFG: sum frequency generation; BS: beam sampler; and GD: glass diffuser.

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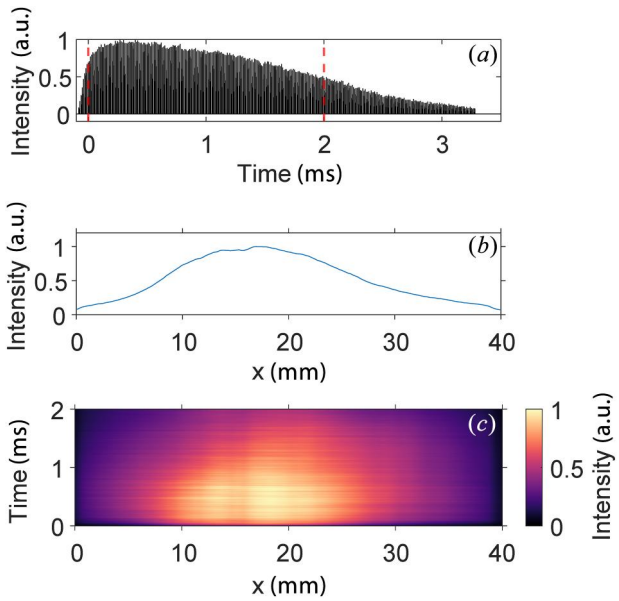

Fig. 3 Time history of OH-PLIF laser pulse energy (*a*), representative laser sheet profile (*b*), and time evolution of sheet profile (*c*)

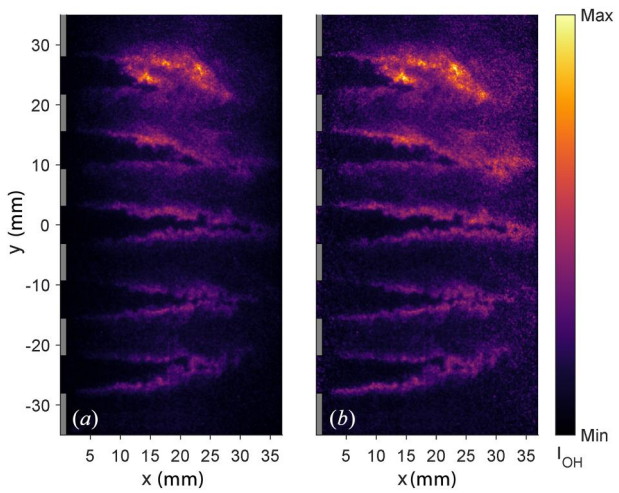

Fig. 4 Original OH-PLIF image (*a*) and corresponding image corrected for sheet nonuniformity (*b*)

throughout the burst, only images from the first 2 ms of the laser burst are analyzed in the present study. The vertical dashed lines in Fig. 3(*a*) denote the beginning and end of the representative test case. Pixel columns are summed for each laser sheet image to produce an intensity distribution over the length of the sheet, which is depicted in Fig. 3(*b*) for a representative sheet image. All profiles from a single 2 ms laser burst are also plotted in Fig. 3(*c*), where slight variations in profile and intensity can be observed over the burst duration.

Characterization of spatial and shot-to-shot nonuniformity within the laser sheet profile is shown in Fig. 3. The sheet profile measurements allow for correction of the original images, as PLIF signal scales linearly with laser excitation energy. The 1D intensity profiles are normalized, scaled, and interpolated to match the resolution of the PLIF images. Pixel columns of the PLIF images are then divided by the normalized intensity values. To account for laser pulse energy decay over the imaging duration, individual frames are also divided by their corresponding integrated sheet intensity values, which are normalized by the maximum value for each test. A representative original OH-PLIF image is plotted alongside the corrected image in Fig. 4. The original image in Fig. 4(*a*) shows low intensity near the burner face, but features in the jet shear layers are restored in the corrected image in Fig. 4(*b*). Lower intensity regions downstream of the reactant jets are also amplified, yielding a more spatially uniform signal. Although this method of correction is accompanied with a discernible increase in noise, a more faithful interpretation of the images is possible as a result.

3 Results and Discussion

A summary of cases examined in this work is provided in Table 1. Three of the cases correspond to H_2/CH_4 fuel mixtures and are denoted by prefix M. The other cases consist of $H_2/NH_3/CH_4$ ternary mixtures and are designated with prefix A. Each designation consists of a 2 L dominant case (2) and a 1 L dominant case (3), while case M-1 represents a low p'_{RMS} case. A constant air flowrate of 0.454 kg/s ($\pm$0.005 kg/s) is maintained at an inlet temperature of 758 K ($\pm$3 K) for all cases. A mean combustion chamber pressure (p_{CC}) of 10.9 bar ($\pm$0.6 bar) is measured. Equivalence ratios were determined to provide a fixed equilibrium flame temperature (T_{ad}) of 1940 K, to match the turbine inlet temperature of >60% combined cycle efficiency gas turbine combustors [33,34].

A representative pressure time history from measurements at location C2 is depicted in Fig. 5(*a*). The 100 ms region of stationary dynamics is indicated by the shaded region and is verified against a series of unit root tests [35,36]. Discrete Fourier transforms of the stationary pressure data are computed, and the spectral power distribution is shown for all conditions in Figs. 5(*b*) and 5(*c*). Figure 5(*b*) corresponds to H_2/CH_4 fuel mixtures (M). Across these cases, an $\sim$1100 Hz mode is observed. Analysis of the natural acoustic frequencies of the combustor is used to verify that this resonance corresponds to the 2 L mode of the combustor [13]. Instability amplitude is observed to increase with methane addition. At $\chi = 0.31$, the combustor transitions to a $\sim$550 Hz mode, which corresponds to the fundamental 1 L mode of the combustor. The spectral power distribution is also depicted for $H_2/NH_3/CH_4$ (A) cases in Fig. 5(*c*). Once again, the 2 L and 1 L modes of the combustor are verified in both cases, with the A-2 case showing 2 L prominence while the A-3 case depicts stronger excitation of the 1 L mode. The observed sensitivity of combustion dynamics to fuel composition, as well as the dominant thermo-acoustic mode transition, highlights the importance of spatiotemporally resolved heat release measurements in the flame. The acquisition of 100 kHz

Table 1 Summary of all test conditions

Case	$\dot{m}_{air}$ (kg/s)	T_{IN} (K)	p_{CC} (bar)	T_{ad} (K)	χ	η	ϕ	p'_{rms}/p_{CC} (%)	Le_{eff}	$\dot{m}_{NH_3}$ (g/s)	$\dot{m}_{CH_4}$ (g/s)	$\dot{m}_{H_2}$ (g/s)	$\dot{m}_{N_2}$ (g/s)
M-1	0.455	759	11.2	1934	0.90	1.00	0.54	0.50	0.71	0.0	3.19	5.57	24.88
M-2	0.454	760	10.4	1917	0.41	1.00	0.55	6.64	0.77	0.0	11.5	1.48	6.49
M-3	0.452	760	10.4	1948	0.31	1.00	0.57	9.17	0.81	0.0	12.81	1.08	5.13
A-2	0.455	757	10.9	1931	0.80	0.61	0.57	3.90	0.77	9.89	5.89	2.79	12.15
A-3	0.459	758	11.0	1936	0.80	0.42	0.58	8.15	0.81	15.56	6.09	1.97	9.01

The hydrogen fraction (χ) and ammonia decomposition efficiency (η) describe the composition of fuel per the expression $\chi[\eta((3/2)H_2 + (1/2)N_2) + (1-\eta)NH_3] + (1-\chi)CH_4$.

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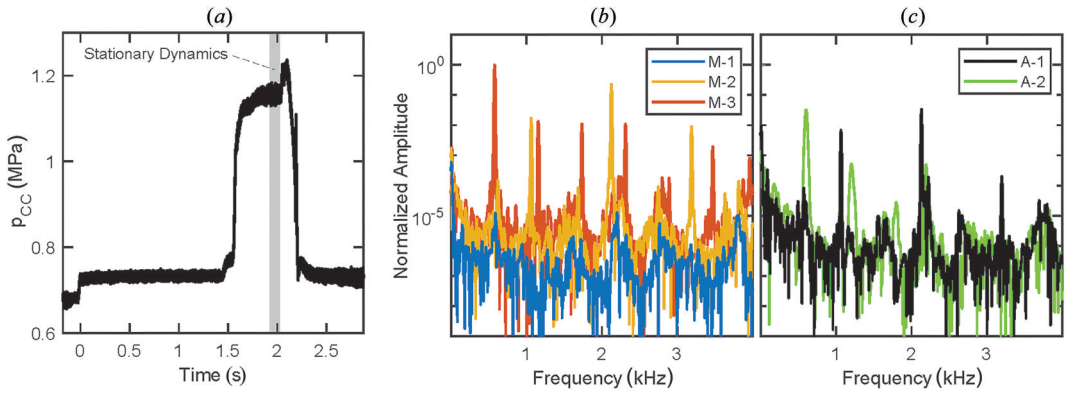

Fig. 5 Representative pressure time history (*a*) and spectral power distribution (*b*) and (*c*) from pressure measurements at location C2

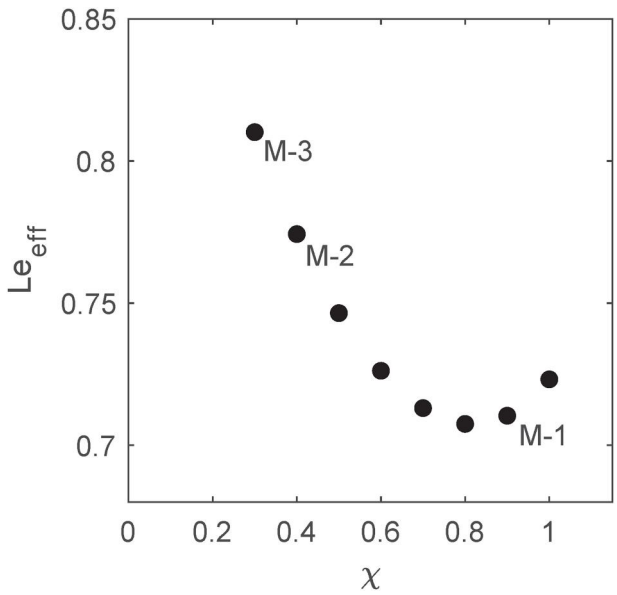

Fig. 6 Variation of effective Lewis number with χ

OH PLIF images enables the analysis of changes to pertinent flow–flame interactions with fuel composition.

3.1 Effect of Methane Addition. Within the flamelet regime, hydrogen enrichment of methane–air flames enhances turbulent burning velocity for consistent turbulence intensity [18,37]. For flames with an effective Lewis number less than unity, the flame front is governed by molecular diffusion, and perturbations lead to increased wrinkling through thermodiffusive instabilities. The increased flame surface area leads to increased turbulent burning velocity. To quantify this effect, a volume weighted effective Lewis number (Le_{eff}) is computed for all $\eta = 1.0$ fuel blends between $\chi = 0.3$ and $\chi = 1.0$ in 0.1 intervals [7]. Results are depicted in Fig. 6. The derivative of Le_{eff} is observed to increase as χ decreases, leading to larger changes in turbulent burning velocity and flame stretch sensitivity as hydrogen is substituted for natural gas.

Instantaneous OH-PLIF images for the M-1 case are shown in Fig. 7(*a*) alongside synchronized p' measurements at location C2 in Fig. 7(*b*). This case corresponds to the most hydrogen prominent fuel mixture of the methane (M) cases. Low amplitude p' fluctuations are observed. Attenuation of the OH signal is observed in the $-y$ direction for each image, which is a result of laser sheet absorption through the probe volume. Regions of significant OH intensity are compact and localized around the reactant jets, indicating flame

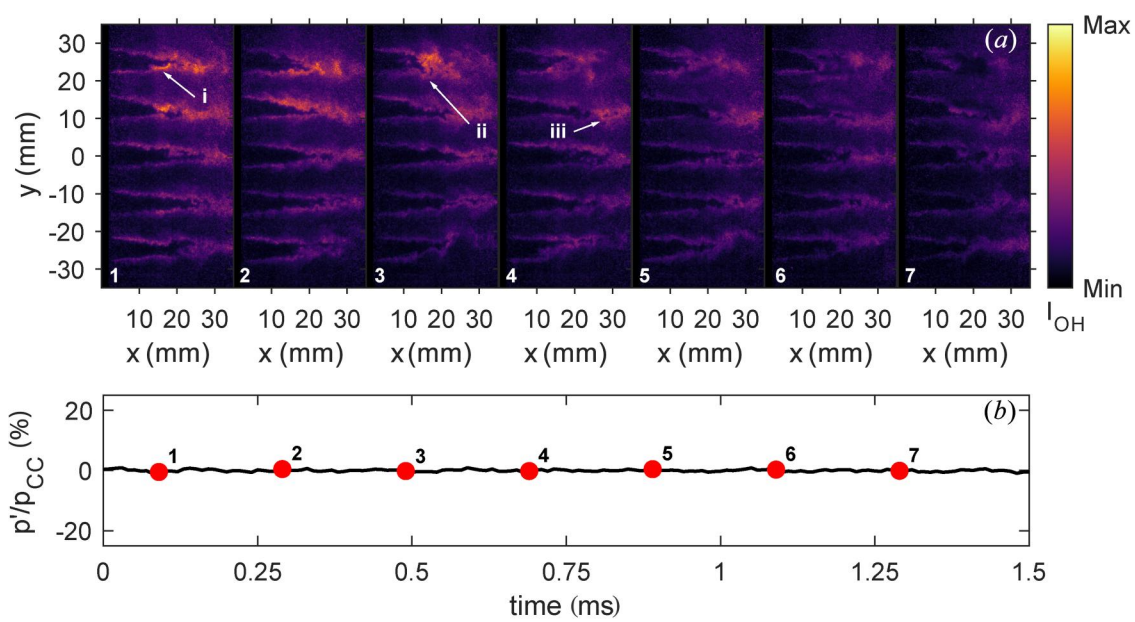

Fig. 7 Instantaneous OH-PLIF snapshots (*a*) and corresponding p' time history (*b*) depicted for case M-1

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stabilization within the shear layer (i). The highly reactive fuel mixture produces a thin reaction zone [38], and regions between the jets show a void in OH signal intensity indicating a lack of reactions occurring in these areas. Significant flame wrinkling is observed, resulting in fine scale flame structures surrounding the premixed jets. The scale of turbulent flame structures increases as the flow is advected, with larger features observed at the end of the reactant jets (ii). Unburned pockets of reactant are periodically shed from the ends of the jets, leading to local surface area increase and subsequent burning that is highlighted through regions of higher signal intensity (iii). While fluctuations in the flame surface area result in unsteady heat release, they are largely induced by hydrodynamic fluctuations of the reactant jet. The stretch sensitive fuel composition leads to rapid response of the flame to these disturbances [25]. The heat release fluctuations observed in this case are largely governed by the length-scales and timescales of the cascading turbulent structures present [28]. This wide range of timescales broadens the response of the flame in frequency space [26], resulting in weak coherence to the natural acoustic modes of the combustor.

A similar set of simultaneous OH-PLIF and p' measurements are shown for the M-2 case in Fig. 8. In contrast to Fig. 7(*b*), p' fluctuations as large as 10% of the mean chamber pressure are observed. The largest contribution to this amplitude occurs at a frequency consistent with the 2 L acoustic mode of the combustion chamber, which corresponds to a full-wave mode shape. Assuming small acoustic disturbances, low mean flow Mach number, and uniform gas composition, a positive pressure antinode at one end of a closed duct will result in the same antinode at the other end for this mode shape. Thus, p' measurements at C2 will be closely in-phase with pressure fluctuations in the flame zone. Selection of instantaneous images represented in Fig. 8(*a*) was determined to highlight different phases of the acoustic cycle. Snapshots are distributed across a single acoustic period with uniform 100 μs spacing for all frames except the first. Strong fluctuations in OH intensity are observed across the acoustic period. Large changes to the length and diameter of the premixed jets are accompanied with significant distortion of the flame front. The first frame occurs at a local minimum in p' and depicts reactant jets that are elongated from the expansion of the acoustic wave (i). This is followed by rapid burning and a reduction in jet length during the compression phase of the cycle on frames 2 and 3 (ii). The recovery of reactant jets is observed from frame 4 to frame 7 as the expansion phase of the next cycle begins (iii).

Throughout the cycle, regions of high OH intensity are observed in the regions between reactant jets (iv). The interactions between jets and resulting recirculation zones are able to sustain combustion for this fuel mixture, which provides a source of ignition for incoming reactants. This observation is distinct from the depicted images of the M-1 case. The substitution from hydrogen to methane alters several aspects of flame propagation that coincide with the highlighted differences. The reduced effective Lewis number indicates a reduction in thermodiffusive effects that contribute to local flame speed [12]. The subsequent effect of flame curvature annihilation must also be considered. As flame propagation occurs normal to the surface, the flame front is smoothed through kinematic restoration [39]. The timescale of this restoration and subsequent heat release is sensitive to variations in local flame speed caused by a decrease in hydrogen. By reducing the effects of preferential thermodiffusive instabilities and flame curvature reduction through kinematic restoration, the frequency response of the flame is narrowed [25,26]. Unsteady heat release is thus more coherent with the timescales of acoustic pressure disturbances, and the 2 L mode of the combustor is excited.

Figure 9 shows a set of OH-PLIF snapshots and coinciding p' measurements for case M-3. The amplitude of pressure fluctuations is notably higher for this case than M-2. Unlike the M-2 case, the dominant frequency corresponds to a half-wave mode shape, and pressure fluctuations in the flame zone will have a phase offset near 180 deg compared to the measurements at the C2 location [13]. The original C2 measurements are plotted alongside a y-axis reflection (labeled p'_{FOV}) in Fig. 9(*b*) to more closely resemble the expected pressure fluctuations at the flame. Instantaneous images in Fig. 9(*a*) are again selected to span a single acoustic cycle and are distributed equally with 200 μs separation. Global oscillations in OH intensity are once again observed, but with several key distinctions. The expansion of the acoustic cycle, beginning on frame 2, corresponds to a sharp increase in bulk velocity which draws unburned reactants into the combustor. Growth in the extent and diameter of the premixed jets is observed in frame 3 and frame 4, as bulk velocity continues to increase (i). Gradients in OH intensity around the jets also reduce, indicating a reduction in reaction progress rate occurring with the acceleration of the flow. At a local minimum in p', the fluctuating velocity reaches a maximum. Unburned reactants fill the entire field of view and suppress reactions, leading to flame pinch-off (ii). As velocity decreases during compression, reactants are rapidly consumed and OH intensity increases again. This process

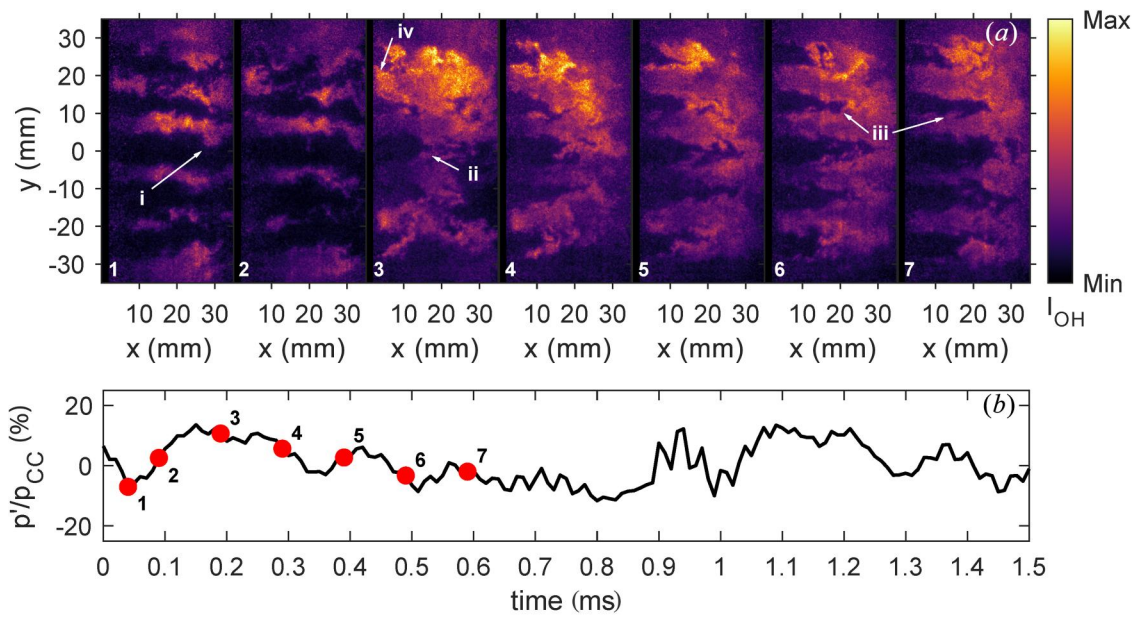

Fig. 8 Instantaneous OH-PLIF snapshots (*a*) and corresponding p' time history (*b*) depicted for case M-2

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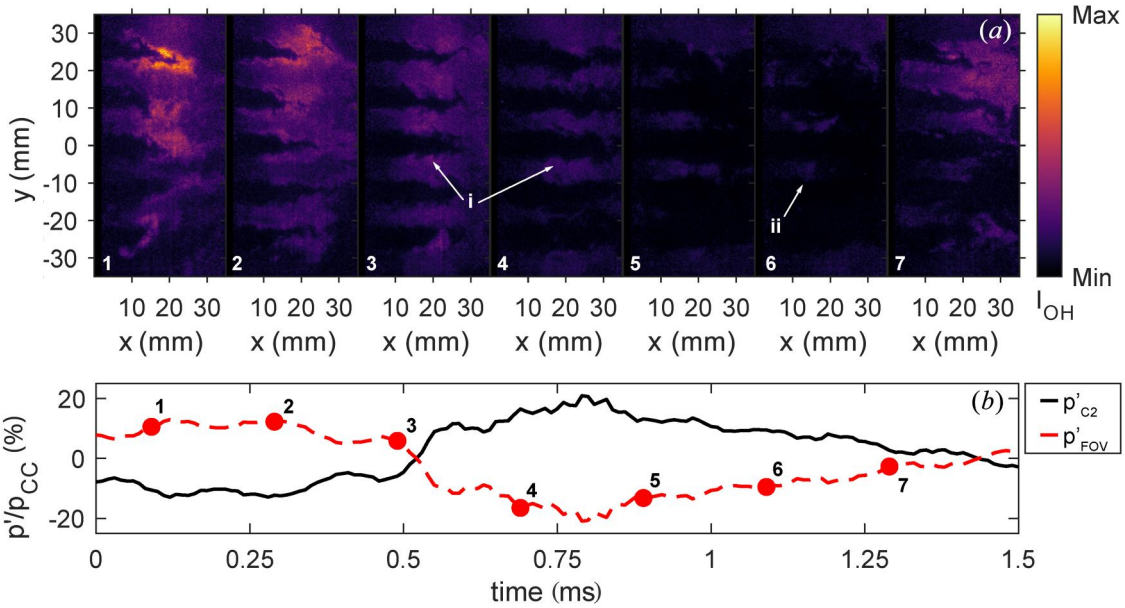

Fig. 9 Instantaneous OH-PLIF snapshots (*a*) and corresponding p' time history (*b*) depicted for case M-3

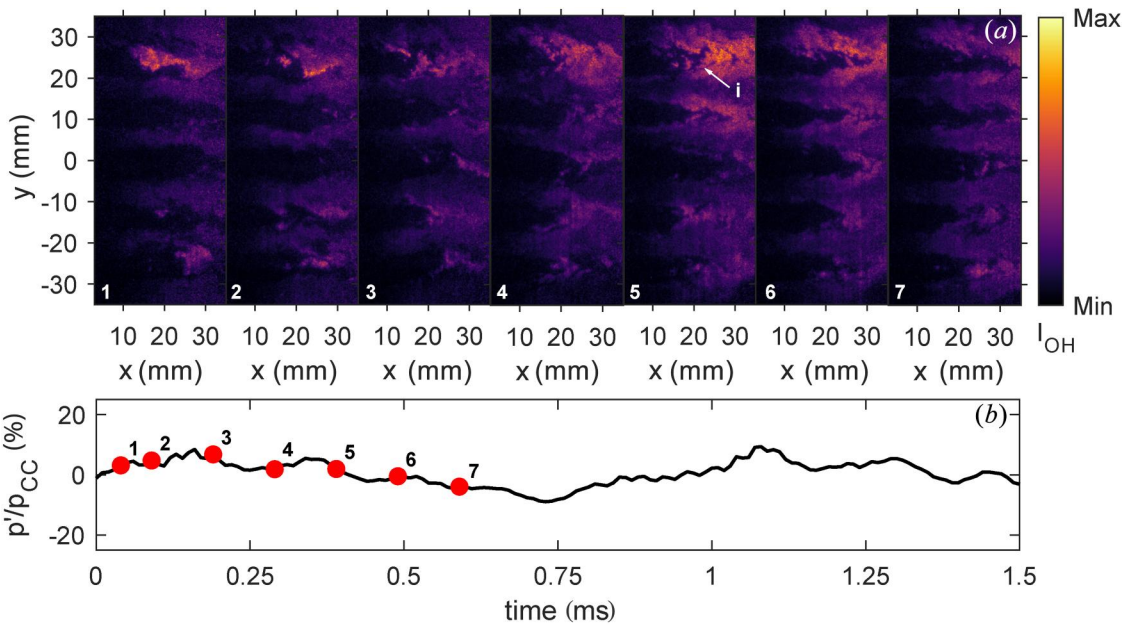

Fig. 10 Instantaneous OH-PLIF snapshots (*a*) and corresponding p' time history (*b*) depicted for case A-2

results in coherent heat release at a frequency resonant with the fundamental 1 L mode of the combustor.

The distinct pinch-off behavior of the M-3 case is attributed to a further reduction in reactivity and stretch sensitivity compared to the other CH_4 cases. As previously noted, the largest change in Le_{eff} occurs between the M-2 and M-3 cases in Fig. 6. The increasing sensitivity of Lewis number with methane addition highlights the different flame responses observed as fuel composition varies.

3.2 Effect of Ammonia Addition. While the observed changes in unsteady heat release can be linked to underlying physical mechanisms governing the propagation of H_2/CH_4 flames, the effect of ammonia addition in a ternary blend is much less understood. To investigate the impact of ammonia addition, OH-PLIF images are generated for the A-2 and A-3 cases and depicted in Figs. 10 and 11, respectively. The distributions of OH in these cases are notably weaker than those of Figs. 8 and 9, which suggest a more distributed reaction zone with the addition of ammonia. Indeed, ammonia flames are typically characterized by low reactivity and diffuse heat release [29].

The flame response to the 2 L mode, depicted in Fig. 10(*a*), shows longitudinal fluctuations of the reactant jets consistent with the compression and expansion phases of the acoustic cycle. Oscillations are notably less than the M-2 case depicted in Fig. 8(*a*), which is likely due to the difference in p'_{RMS} between the two. Despite this, key differences in flame structure and evolution are observed. The largest gradients in OH signal occur at the ends of the jets and are characterized by regions of significant flame corrugation, as seen in the uppermost jet on frames 3–6 (i). This contrasts the high intensity regions of Fig. 8(*a*) characterized by rapid burning through increases in flame surface area. The reduction in kinetic rates with the addition

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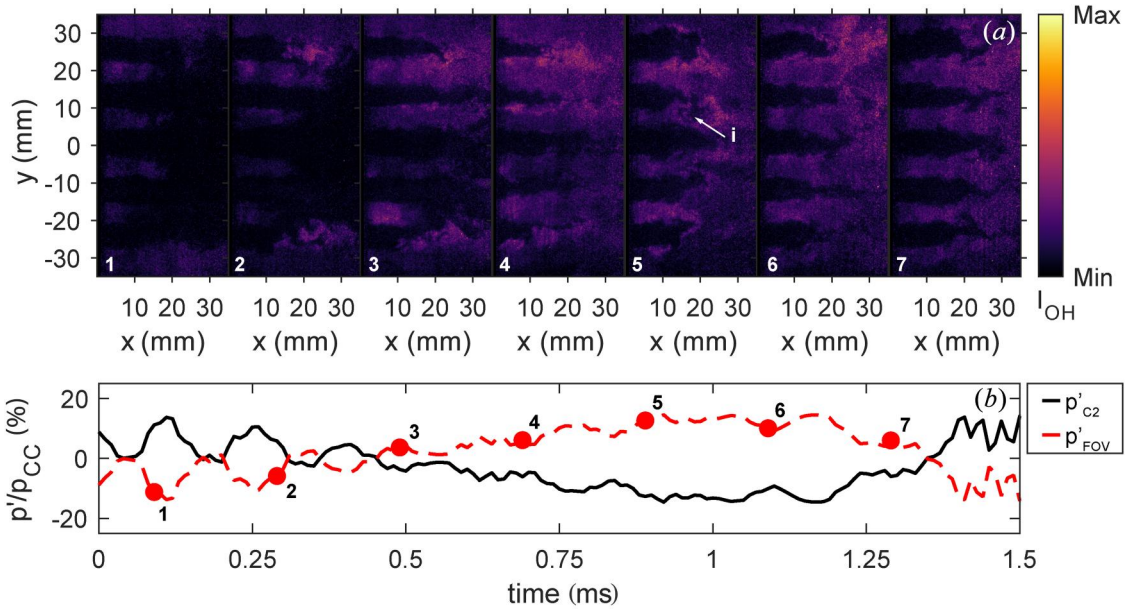

Fig. 11 Instantaneous OH-PLIF snapshots (*a*) and corresponding p' time history (*b*) depicted for case A-3

of ammonia results in longer timescales for preheat and consumption of reactants. This increases the flame thickness and thus the length-scales of flame–turbulence interaction.

Figure 11 portrays several OH-PLIF snapshots with simultaneous p' measurements for the A-3 case. The global response of the flame is similar to the results seen in Fig. 9(*a*), but several unique features are noted. During compression on frames 3–5, reactant jets near the walls of the combustor appear shorter than the center jet. A flame–vortex interaction is also observed on frame 5 (i), showing entrainment of reactants and enhanced burning in the surrounding region. The central jet at $y = 0$, conversely, appears straighter with less large-scale fluctuations in the surrounding flame surface area. It is possible that the symmetry of the central jet leads to a relative balance of flame–flame interactions that results in a reduced impact on the local flame structure.

4 Conclusions

The impact of fuel composition on the structure and dynamics of lean-premixed H_2/CH_4 and $H_2/NH_3/CH_4$ flames was investigated in a high-pressure multi-element combustor. Time resolved images of the flame response to acoustic disturbances were obtained using 100 kHz OH-PLIF along with simultaneous measurements of chamber pressure fluctuations. Two unstable modes of the combustor were studied with and without the inclusion of ammonia, as well as a case of stable operation. The following conclusions are noted:

(1) The increase in instability magnitude for test cases with methane addition can be correlated to increased corrugation of the flame front and to commensurate heat release fluctuations. Variations in flame speed, stretch sensitivity, and thermodiffusive effects with fuel composition alter the timescales of unsteady heat release, allowing in-phase coupling with the natural acoustic modes of the combustor.
(2) Unstable cases corresponding to the 2 L mode of the combustor see regions of OH between reactant jets persisting through the expansion phase of the acoustic cycle. In contrast, flame pinch-off, extinction, and re-ignition are seen globally in cases that excite the 1 L mode.
(3) Ammonia addition results in weaker gradients in OH intensity, which indicates a greater dispersion of reactions and a corresponding increase in flame thickness. The resulting flame structure sees significant corrugation as a result.

These findings highlight the disparate chemical and mixing timescales of ammonia, hydrogen, and methane combustion and provide understanding of the resulting effects on the structure and dynamics of these flames in turbulent, high-pressure environments. For retrofit gas turbine combustor applications, it is clear that significant consideration of the injector geometry is necessary for fuel flexible operation with these fuels. However, the unique properties of premixed hydrogen flames could be leveraged for flame stability with ammonia and methane in these devices.

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Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

$\mathrm{Le}_{\mathrm{eff}}$ = effective Lewis number
$\dot{m}_{\mathrm{air}}$ = combustor air mass flow rate, kg/s
p_{CC} = chamber pressure, bar
p'_{RMS} = root-mean-square of pressure fluctuation, kPa
T_{AD} = computed equilibrium flame temperature, K
T_{IN} = inlet air temperature, K
η = efficiency of ammonia decomposition to hydrogen

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ϕ = equivalence ratio
χ = hydrogen fraction

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